

WEEK 5



First Pyramid

Stack shrinking squares.

첫 피라미드 — 점점 작아지는 층 쌓기 · 영어코딩

BIG PICTURE

Layers that shrink ▼



Each layer is a square, 2 smaller than the one below.

한 층씩 2칸 작아지며 위로

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

Why shrink by 2?

A pyramid narrows on **both** sides each layer — left and right.

Lose 1 block each side = $\text{size} - 2$ per layer. That keeps it centered.

양쪽에서 1칸씩 → 2칸 줄어듭니다

CONCEPT

The `while` loop

```
while size > 0:  
    build_layer(size)  
    agent.move(UP, 1)  
    size = size - 2
```

`while` repeats **as long as** the condition is true.

`while` = 조건이 참인 동안 반복

WATCH IT!

size counts down, then exits

```
size = 9
```

```
size > 0
```

```
build_layer(size); move up; size -= 2
```

```
▸ layers built
```

CODE

The full pyramid

```
size = 9

while size > 0:
    for row in range(size):
        for col in range(size):
            agent.place(DOWN)
            agent.move(FORWARD, 1)
            agent.turn(RIGHT)
        agent.move(UP, 1)
    size = size - 2
```


PREDICT 🤖

List the layer sizes

Predict first — then click reveal

```
size = 9
while size > 0:
    build_layer(size)
    agent.move(UP, 1)
    size = size - 2
```

Reveal answer 🔍 reveal

VOCAB

New words today

while	repeat while a condition stays true
layer	one square level of the pyramid
condition	the true/false test that controls the loop
shrink	get smaller each layer (size – 2)

~하는 동안

층

조건

줄어들다

 DEBUG

What's wrong here?

buggy

```
# each layer should be 2 smaller  
size = 9  
while size > 0:  
    build_layer(size)  
    agent.move(UP, 1)  
    size = size - 1
```

Think first — then click next.



DEBUG

The fix

Bug: `size - 1` shrinks only one side, so the layers are not centered and it looks lopsided. A real pyramid loses a block on *each* side: `size - 2`.

fixed

```
size = size - 2
```

YOUR TURN!

Build a size-7 pyramid ▲

- ✓ Start with `size = 7`
- ✓ Use a `while` loop, `size - 2` each layer
- ✓ Move up 1 between layers ($7 \rightarrow 5 \rightarrow 3 \rightarrow 1$)
- ✓ Build it again in a different material
- ✓ Send a photo + walkthrough on KakaoTalk



QUIZ

QUESTION 1 OF 3

Why does the pyramid shrink by 2 each layer?

It narrows on both sides — one block each side.

To make it taller.

2 is faster to type.

Minecraft requires even numbers.



QUIZ

QUESTION 2 OF 3

What's wrong with `size = size - 1` here?

Nothing — it's correct.

The loop never ends.

size can't go below 1.

It shrinks only one side — the layers aren't centered.



QUIZ

QUESTION 3 OF 3

Starting at `size = 9`, how many layers?

9 layers.

5 layers (9, 7, 5, 3, 1).

4 layers.

Infinite.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



First pyramid built!

Next: flip it — build an inverted pyramid.

다음 주: 거꾸로 된 피라미드 만들기